11. TCP vs UDP, Handshakes, and Reliability

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Phase: Phase 1 — Models and packets
Estimated time: 1–2 days
Slug: `11-tcp-udp`

Learning objectives

- Contrast TCP and UDP properties
- Explain three-way handshake conceptually
- Know when UDP is preferred
- Observe TCP with ss/tcpdump

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

TCP vs UDP

	TCP	UDP
Connection	yes	no
Reliable delivery	yes	best-effort
Ordered	yes	no
Use cases	HTTP, SSH	DNS, QUIC/games, media

```
# Established TCP
ss -tan state established | head
# Capture handshake (lab)
sudo tcpdump -ni any -c 20 'tcp[tcpflags] & (tcp-syn) != 0'
# UDP DNS
sudo tcpdump -ni any -c 5 udp port 53
```

SYN → SYN-ACK → ACK establishes TCP. FIN/RST close or reset.

Common mistakes

- Using UDP and expecting TCP guarantees
- SYN floods without rate limits
- Forcing TCP where UDP+app reliability fits

Exercises

Work in `exercises/11-tcp-udp/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/11-tcp-udp/` until after an honest attempt.

Self-check

1. Three-way handshake steps?
2. Is DNS always UDP?
3. Command to show established TCP?

Further reading

- https://www.rfc-editor.org/rfc/rfc9293